NIVRA

Before impact.

Bal Bharati Hackathon 3.0 / Idea Submission

PROBLEM STATEMENT

Developing incidents can go unnoticed; camera footage alone cannot bring help.

Theme: T3 — AI for Good

Team: Damage Control

School: Bal Bharati Public School, Pitampura

Team leader: _____

Team size (excluding mentor): ____ Category: ____



Your Idea

AI-assisted early intervention, with a human in charge.





Idea / Solution

Use a phone or camera feed to flag visible movement patterns that may need attention.

Problem Resolution

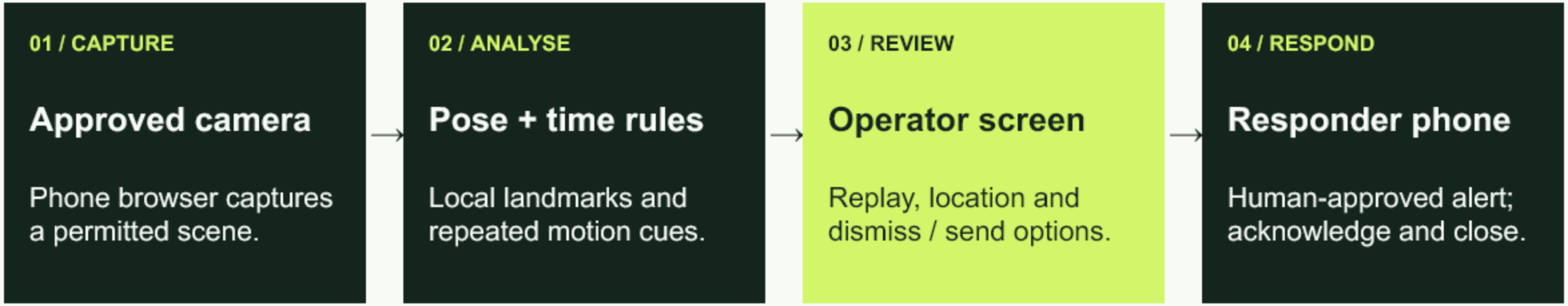
Give staff a short replay and location, then connect a confirmed alert to a responder.

Observe → flag → human review

Decide → dispatch → acknowledge

Technical Approach

Proposed architecture / one supervised demo



Frontend / browser

React views for camera, operator and responder. Run pose inference in a worker.

Backend / event workflow

FastAPI + WebSockets for alerts and acknowledgements; local demo, no cloud required.

AI prioritises review. It does not infer identity, intent or guilt.

Feasibility and Viability

Technical

Existing pose models are available. Reliable escalation cues remain unproven.

Operational

One camera, one operator, one responder. Trial the workflow with school staff.

Resources needed

Two phones, a laptop, network, web/ML skills and mentor support. Build in small stages.

Data & build plan

Permissioned staged clips; none collected yet. Connect → flag → review → acknowledge.

Challenge / risk	Proposed mitigation
Ordinary play creates false alerts	Use temporal rules; require human review before dispatch.
Occlusion, poor light or network loss	Show confidence / offline state; test varied scenes.
Video privacy and child data	Minimise retention and access; obtain school approval and review applicable rules.

Impact and Benefits

More time to respond.

Social

Aim: earlier human attention to developing incidents.

Economic

Explore reuse of existing devices; costs and savings are unvalidated.

Environmental

Possible device reuse; no measured environmental benefit claimed.

End users

Staff receive context, location and a clear response action.

Institutions / industry

A proposed alert-to-acknowledgement record for safety teams.

Scalability

Start with one campus; expand only after controlled testing.

Research and References

Evidence informs the idea; it does not validate NIVRA.

01 / SCHOOL SAFETY

MoE, School Safety & Security Guidelines (2021).
Establishes institutional safety responsibilities.

02 / TECHNICAL BASIS

Google Research, BlazePose (2020), and MediaPipe docs.
Support pose estimation—not prediction of violence.

03 / EXISTING SOLUTION

Axis Object Analytics: object tracking and event triggers.
NIVRA proposes human review plus response acknowledgement.

04 / DATA & PRIVACY

Planned data: permissioned staged volunteer videos; none yet.
DPDP Act (2023): assess applicable privacy rules before a pilot.

Proposed stack

React + browser camera

Three device roles and scene capture.

MediaPipe Pose

Body landmarks in a browser worker.

Temporal rules

Repeated visible changes trigger review.

FastAPI + WebSockets

Alert delivery and acknowledgement.

Team Member Details

Bal Bharati Public School, Pitampura

Students

SL	Student ID	Name	Class	DOB	Email	Phone Number	Gender
1							
2							
3							
4							
5							

Mentor

SL	Name	UID	Address	DOB	Email	Phone Number	Gender
1							